Prop.: $T: V \to V$ is diagonalizable

(判断T是否diag.-able)

$\Downarrow$ (后证前)

$m_T(x) = (x-\mu_1)\cdots(x-\mu_k)$, μ_i's are distinct

(e.g. $m_T(x) = (x-1)^2$ → not distinct → T is not diag.-able

$m_T(x) = (x-1)$ → distinct → T is diag.-able)

Proof:

Let B be a basis of V, s.t. $(T)_{BB} = \begin{pmatrix} \mu_1 & & 0 \\ & \ddots & \\ 0 & & \mu_k \end{pmatrix}$

① $(T_{BB} - \mu_1 I_{n\times n})\cdots(T_{BB} - \mu_k I_{n\times n}) = 0_{n\times n}$

Ex.

Let $p(x) = (x-\mu_1)\cdots(x-\mu_k)$

Then $p(T)_{BB} = ① = 0_{n\times n}$

$\Rightarrow p(T) = 0_{V\to V}$

$\Rightarrow m_T(x) \mid p(x)$

② If you omit any factor $(x-\mu_i)$ of $p(x)$,

(omit: 忽略; 遗漏; 排除)

e.g. $q(x) = (x-\mu_1)\cdots(x-\mu_{i-1})(x-\mu_{i+1})\cdots(x-\mu_k)$

? ② Then $q(T)_{BB} \neq 0_{n\times n}$

$\Rightarrow q(x) \neq m_T(x)$

$\Rightarrow m_T(x) = p(x)$

Primary Decomposition Thm. (PDT)

(② and Diagonalizable Thm.)

(idea: "Decomp." T into $T|_{V_1}: V_1 \to V_1, \cdots, T|_{V_k}: V_k \to V_k$)

$T: V \to V$ a linear operator, $\dim(V) < \infty$,

$m_T(x) = [p_1(x)]^{e_1} \cdots [p_k(x)]^{e_k}$,

Let $V_i = \ker(p_i(T)^{e_i})$

$\Rightarrow$ then: 1. $\forall i$, V_i is T-invariant (i.e., $T(V_i) \leq V_i$

2. $V = V_1 \oplus \cdots \oplus V_k$

3. $T|_{V_i}$ has $m_T(x) = p_i(x)^{e_i}$

Proof:

Consider $q_i(x) := \frac{m_T(x)}{p_i(x)^{e_i}} = p_1(x)^{e_1} \cdots p_{i-1}(x)^{e_{i-1}} p_{i+1}(x)^{e_{i+1}} \cdots p_k(x)^{e_k}$

① $\gcd(q_1(x), \cdots, q_k(x) = 1$

(Bezout) $\Rightarrow a_1(x)q_1(x) + \cdots + a_k(x)q_k(x) = 1$ (for polys. $a_i(x)$) ①

② $\gcd(q_i(x), p_i(x)^{e_i}) = 1$

(Bezout) $\Rightarrow b_i(x)q_i(x) + c_i(x)p_i(x)^{e_i} = 1$

③ $q_i(x)p_i(x)^{e_i} = m_T(x)$

④ If $i \neq j$, $m_T(x) \mid q_i(x) \cdot q_j(x)$

Claim 1: $\ker(p_i(T)^{e_i})$ $(\neq V_i)$ is T-invariant ?

Proof:

Let $\underline{v} \in \ker(p_i(T)^{e_i})$,

then $p_i(T)^{e_i}(\underline{v}) = \underline{0}$

Consider $p_i(T)^{e_i}(T(\underline{v}))$

$$\overset{(*)}{=} T[p_i(T)^{e_i}(\underline{v})] = T(\underline{0}) = \underline{0}$$

$\Rightarrow T(\underline{v}) \in \ker(p_i^{e_i})$

?

(Recall: In general, $\forall A, B: V \to V$, $A \cdot B \neq B \cdot A$

However $h(T) = f(T)g(T) = g(T)f(T)$,

where $h(x) = f(x)g(x)$

For $(*)$, take $f(x) = p_i(x)^{e_i}$, $g(x) = x$)

Claim 2:

$V = V_1 + \cdots + V_k$, where $V_i := \ker(p_i(T)^{e_i})$

Proof:

By ①, $a_1(T)q_1(T) + \cdots + a_k(T)q_k(T) = I_{k\times k}$

$\Rightarrow a_1(T)q_1(T)(\underline{v}) + \cdots + a_k(T)q_k(T)(\underline{v}) = I(\underline{v}) = \underline{v}$, $\forall \underline{v} \in V$

Let $\underline{v}_i = a_i(T)q_i(T)(\underline{v})$

$\Rightarrow \underline{v}_1 + \cdots + \underline{v}_k = \underline{v}$

Note: $\underline{v}_i$ in $(*)$ is in V_i

$p_i(T)^{e_i}(a_i(T)q_i(T)(\underline{v})) \overset{\text{(Recall)}}{=} a_i(T)p_i(T)^{e_i}q_i(T)(\underline{v})$

③ $\Rightarrow a_i(T)m_T(T)(\underline{v}) = a_i(T)(\underline{0}) = \underline{0}$

Claim 3:

$$V = V_1 \oplus V_2 \oplus \cdots \oplus V_k$$

Proof:

Suppose $\underline{v}_1' + \cdots + \underline{v}_k' = \underline{0}$,

where $\underline{v}_i' \in V_i = \ker(p_i(T)^{e_i})$

(WTS: $\underline{v}_1' = \cdots = \underline{v}_k' = \underline{0}$)

By ②, $b_i(T)q_i(T) + c_i(T)p_i(T)^{e_i} = I$

$$\Rightarrow b_i(T)q_i(T)(\underline{v}_i') + c_i(T)p_i(T)^{e_i}(\underline{v}_i') = I(\underline{v}_i') = \underline{v}_i'$$

① $\Rightarrow b_i(T)q_i(T)(\underline{v}_i') = \underline{v}_i'$

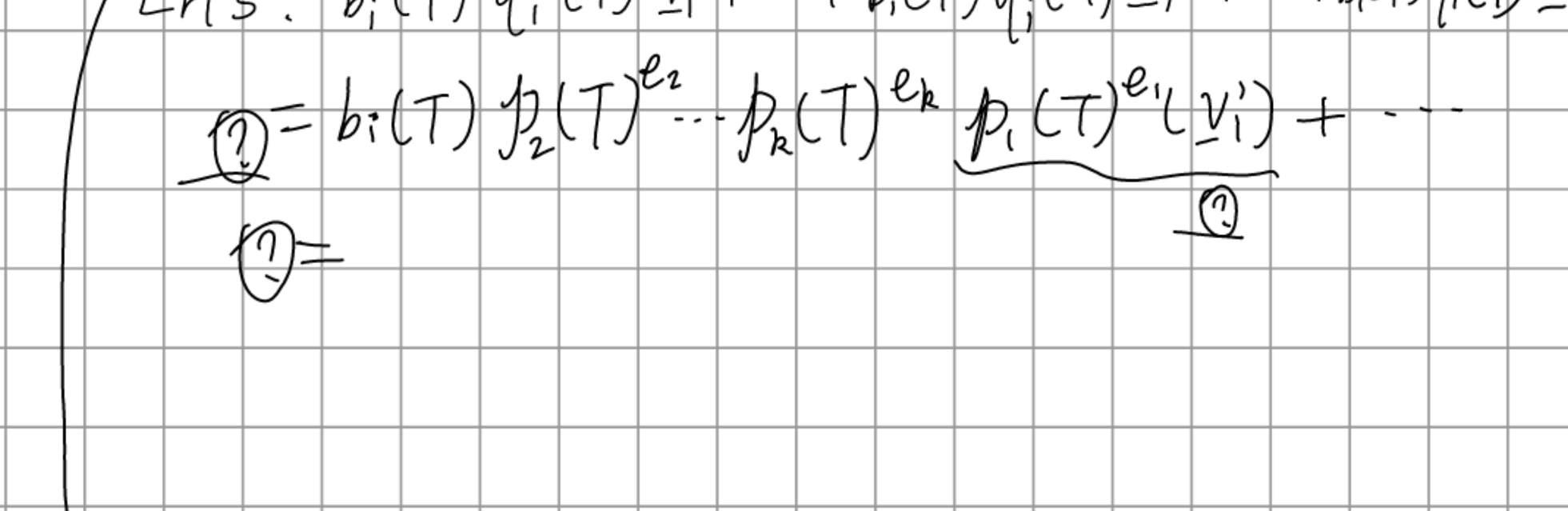
↳ apply to $\underline{v}_1' + \cdots + \underline{v}_k' = \underline{0}$

LHS: $b_i(T)q_i(T)\underline{v}_1' + \cdots + b_i(T)q_i(T)\underline{v}_i' + \cdots + b_i(T)q_i(T)\underline{v}_k'$

① $= b_i(T)p_2(T)^{e_2}\cdots p_k(T)^{e_k}\underbrace{p_1(T)^{e_1}(\underline{v}_1')}_{①} + \cdots$

① $=$

RHS: $b_i(T)q_i(T)(\underline{0}) = \underline{0}$



Claim 4: $T|_{V_i}: V_i \to V_i$
has $m_{T|_{V_i}}(x) = p_i(x)^{e_i}$

Proof: Take $\forall \underline{v}_i \in V_i = \ker(p_i(T)^{e_i})$

$\Rightarrow$ Then $p_i(T)^{e_i}(\underline{v}_i) = \underline{0}$

$\Rightarrow p_i(T|_{V_i})^{e_i}(\underline{v}_i) = \underline{0}$

$\Rightarrow p_i(T|_{V_i})^{e_i}$ is the zero operator

$\Rightarrow m_{T|_{V_i}}(x) \mid p_i(x)^{e_i}$

Suppose on contrary,

$m_{T|_{V_i}}(x) = p_i(x)^{g_i}$, where $g_i < e_i$

$\Rightarrow p_i(T)^{g_i}(\underline{v}) = \underline{0}$, $\forall \underline{v}_i \in V_i$ —(1)

Consider $d(x) := p_1(x)^{e_1} \cdots p_i(x)^{g_i} \cdots p_k(x)^{e_k}$

$\Rightarrow \deg(d(x)) < \deg(m_T(x))$

and $\forall \underline{v} \in V$, $d(T)(\underline{v}) = (p_1(T)^{e_1} \cdots p_k(T)^{e_k}) \cdot (\underline{v}_1 + \cdots + \underline{v}_k)$,

where $\underline{v} = \underline{v}_1 + \cdots + \underline{v}_k$, $\underline{v}_j \in V_j = \ker(p_j(T)^{e_j})$

Then $d(T)(\underline{v}) = \sum_{j=1}^{k} p_1(T)^{e_1} \cdots [p_j(T)^{e_j} \underline{v}_j] \cdots p_k(T)^{e_k}$

$= \underline{0} + \cdots + \underline{0} = \underline{0}$

$\Rightarrow d(T) = 0_{V \to V}$ [$\Rightarrow \deg(d(x)) \geq \deg(m_T(x))$, (1)

which contradicts. Hence $g_i = e_i$

Corollary:

$m_T(x) = (x-\mu_1)\cdots(x-\mu_k)$ μ_i's are distinct

$\Downarrow$

$T: V \to V$ is diagonalizable

Proof:

By PDT, $V = V_1 \oplus \cdots \oplus V_k$,

where $V_i := \ker(T-\mu_i I)$

(take $f_i(x)^{e_i} = (x-\mu_i)$)

$\Rightarrow V_i$ is the μ_i-<u>eigenspace</u> of $T: V \to V$

(?)

$\Rightarrow$ Take basis B_i of V_i

(B_i : basis of μ_i-e.vectors

Then $V = V_1 \oplus \cdots \oplus V_k$

$\Rightarrow B := B_1 \cup \cdots \cup B_k$

is a basis of e.vectors of T

$\Rightarrow T$ is diagonalizable